

- Coulomb Law
- Electric field
- Gauss Law (Maxwell eqn 1)
- Electric field
- capacitors
- Ohm's Law, Circuits, Resistance
- RC circuit
- Magnetic fields, forces
- Induction
- RLC circuits

⇒ openstax.org physics 2

⇒ universite fizigi 2

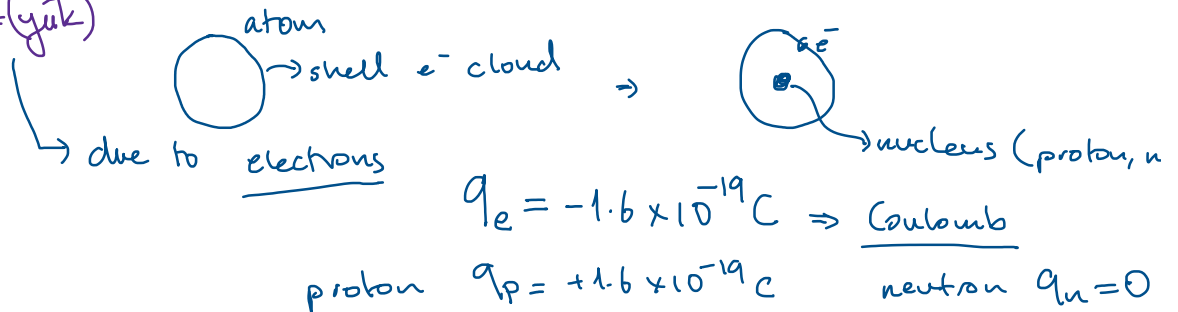
lecture notes

⇒ midterm + final + HW
 ~30% { % HW } program

Electricity & Magnetism

Electrostatic → "stationary"
 ↳ (Lurgun) $\Rightarrow$ Electrodynamics $\Rightarrow$ current

q charge = (yük)



Phys I [kg, m, s] 3v

Phys II [A = Ampere ~ Coulomb/s] 1v

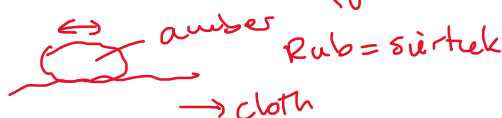
SI units 7 ~ K kelvin

cd candela

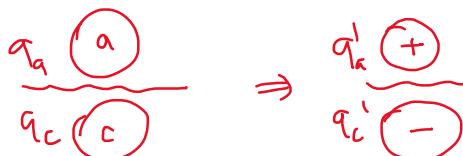
NA Avogadro's no.

charge: $q \sim e^-$

electron = amber = kahrībar (greek)



takip e^- from atom



$q_a + q_c = q'_a + q'_c =$ conservation of charge

$q + \Rightarrow$ (+, +) $\Rightarrow$ lack of e^-

(-, -) $\Rightarrow$ surplus of e^-

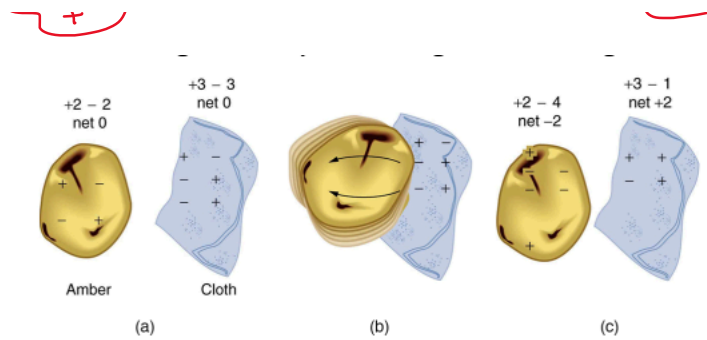


FIGURE 18.8 When materials are rubbed together, charges can be separated, particularly if one material has a greater affinity for electrons than another. (a) Both the amber and cloth are originally neutral, with equal positive and negative charges. Only a tiny fraction of the charges are involved, and only a few of them are shown here. (b) When rubbed together, some negative charge is transferred to the amber, leaving the cloth with a net positive charge. (c) When separated, the amber and cloth now have net charges, but the absolute value of the net positive and negative charges will be equal.

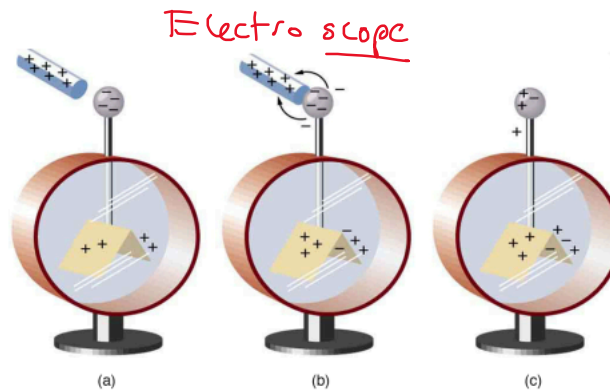
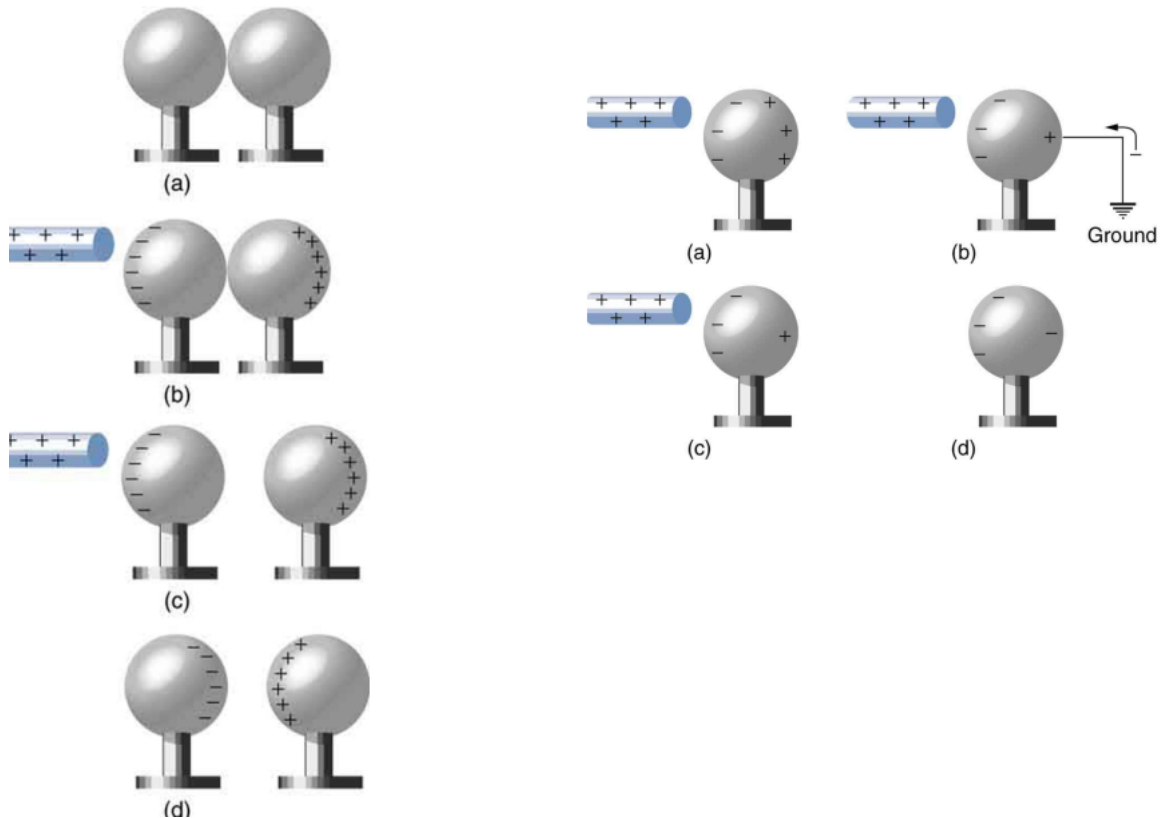
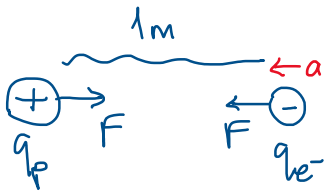


FIGURE 18.11 An electroscope is a favorite instrument in physics demonstrations and student laboratories. It is typically made with gold foil leaves hung from a (conducting) metal stem and is insulated from the room air in a glass-walled container. (a) A positively charged glass rod is brought near the tip of the electroscope, attracting electrons to the top and leaving a net positive charge on the leaves. Like charges in the light flexible gold leaves repel, separating them. (b) When the rod is touched against the ball, electrons are attracted and transferred, reducing the net charge on the glass rod but leaving the electroscope positively charged. (c) The excess charges are evenly distributed in the stem and leaves of the electroscope once the glass rod is removed.



μ_0 = magnetic permeability of space

$$\frac{1}{\sqrt{\mu_0 \epsilon_0}} = c = \text{speed of light}$$



$$F = k \frac{q_p q_e}{r^2} = 9 \times 10^9 \frac{(1.6 \times 10^{-19})^2}{1^2} = 2.3 \times 10^{-28} \text{ N}$$

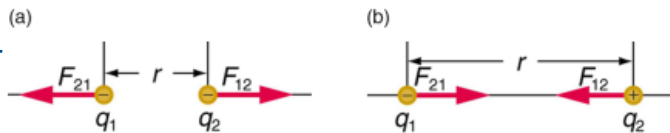
$$F = ma$$

an e^- experiences an acceleration of $30g$ at this situation

$$a = \frac{F}{m_e} = \frac{2.3 \times 10^{-28} \text{ N}}{9 \times 10^{-31} \text{ kg}} = 0.3 \times 10^3 \text{ m/s}^2 = 300 \text{ m/s}^2$$

what's the last time you experience $a = g$?!

Electrostatic (Coulomb's) Force $\vec{F} = k \frac{q_1 q_2}{r^2} \hat{r}$ $\left\{ \hat{r} \text{ is direction drawn from centers of charges} \right.$



$$F = k \frac{|q_1 q_2|}{r^2} \Rightarrow \text{magnitude of } \vec{F}$$

$$k = 8.988 \times 10^9 \frac{\text{N} \cdot \text{m}^2}{\text{C}^2} \approx 8.99 \times 10^9 \frac{\text{N} \cdot \text{m}^2}{\text{C}^2} \approx 9 \times 10^9 \frac{\text{N} \cdot \text{m}^2}{\text{C}^2}$$

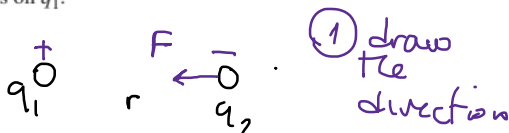
$$\hat{x}, \hat{y} \Rightarrow \hat{r}, \hat{\theta}$$

$$\vec{F} = k \frac{q_1 q_2}{r^2} \hat{r}$$

Two point charges, $q_1 = +25 \text{ nC}$ and $q_2 = -75 \text{ nC}$, are separated by a distance $r = 3.0 \text{ cm}$ (Fig. 21.12a). Find the magnitude and direction of the electric force (a) that q_1 exerts on q_2 and (b) that q_2 exerts on q_1 .

$n = \text{nano}$
 10^{-9}

Two point charges are located on the x-axis of a coordinate system $q_1 = 1.0 \text{ nC}$ is at $x = +2.0 \text{ cm}$, and $q_2 = -3.0 \text{ nC}$ is at $x = +4.0 \text{ cm}$. What is the total electric force exerted by q_1 and q_2 on charge $q_3 = 5.0 \text{ nC}$ at $x = 0$?



② calc. magnitude (besar kecil)

$$F = k \frac{q_1 q_2}{r^2} = |\vec{F}|$$

$$= 9 \times 10^9 \times 25 \times 10^{-9} \times 75 \times 10^{-9} / (0.03)^2$$

$$= \frac{1 \times 25 \times 75 \times 10^{-9}}{10^{-4}} = 1875 \times 10^{-5} \text{ N} \approx 1.9 \times 10^{-2} \text{ N}$$



Force on q_3 ?

$$\vec{F}_{13} \quad \vec{F}_{23} \quad \Sigma \vec{F} = ?$$

$$F_{13} = 9 \times 10^9 \times 10^{-9} \times 5 \times 10^{-9} / 4 \times 10^{-4}$$

$$F_{13} = \frac{45}{4} \times 10^{-5} \text{ N}$$

$$F_{23} = 9 \times 10^9 \times 3 \times 10^{-9} / 16 \times 10^{-4}$$

$$F_{13} \leftarrow \rightarrow F_{23} = 8.44 \times 10^{-5} \text{ N}$$

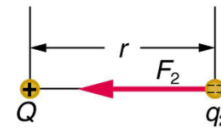
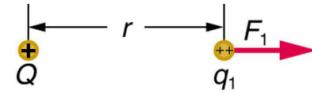
$$11.3 \times 10^{-5} \text{ N}$$

$$\vec{F} = -2.8 \times 10^{-5} \text{ N} ; \quad \Sigma F = 2.8 \times 10^{-5} \text{ N} ;$$

Concept of a Field

A field is a way of conceptualizing and mapping the force that surrounds any object and acts on another object at a distance without apparent physical connection. For example, the gravitational field surrounding the earth (and all other masses) represents the gravitational force that would be experienced if another mass were placed at a given point within the field.

In the same way, the Coulomb force field surrounding any charge extends throughout space. Using Coulomb's law, $F = k|q_1q_2|/r^2$, its magnitude is given by the equation $F = k|qQ|/r^2$, for a **point charge** (a particle having a charge Q) acting on a **test charge** q at a distance r (see Figure 18.18). Both the magnitude and direction of the Coulomb force field depend on Q and the test charge q .



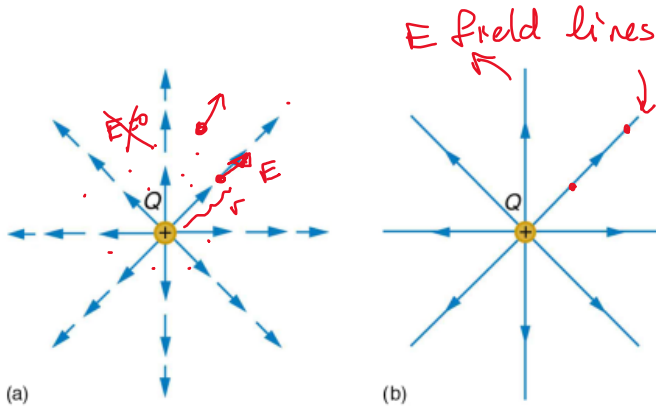
From <https://openstax.org/books/college-physics-2e/pages/18-4-electric-field-concept-of-a-field-revisited>

$$Q \text{ --- } r \text{ --- } q \rightarrow F = \frac{kQq}{r^2}$$

$$Q \text{ --- } r \rightarrow \vec{E} = \frac{\vec{F}}{q} \equiv \text{Electric field}$$

$$Q \rightarrow \vec{E} = \frac{kQ}{r^2} \hat{r}$$

$$E = \left[\frac{N}{C} \right]$$



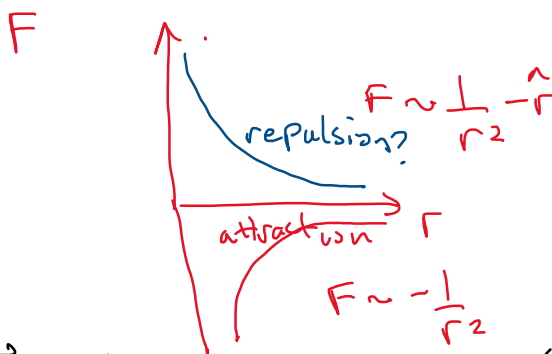
look into applets

$$F = \frac{kq_1q_2}{r^2} \quad r \rightarrow 0 \quad F \rightarrow \infty$$

$$E = \frac{kq}{r^2} \quad r \rightarrow 0 \quad E \rightarrow \infty$$

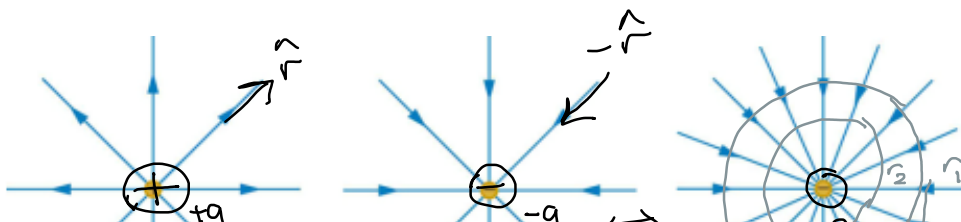


Hydrogen atom = stable?
Pauli exclusion prin

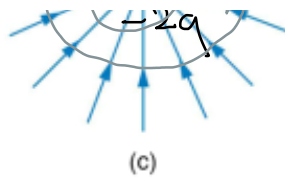
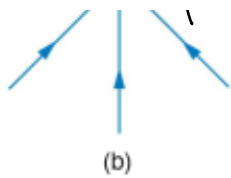
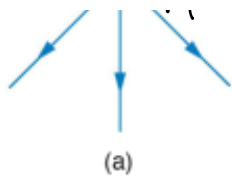


repulsion + attraction
{ nuclear forces }

$$\vec{E} = \frac{1}{q} \frac{\vec{F}}{q} \quad \left\{ \begin{array}{l} \oplus \rightarrow \vec{E} \hat{r} \\ \ominus \leftarrow \vec{E} \hat{r} \end{array} \right\} \quad \vec{E} \equiv \frac{\vec{F}}{q}$$



$\oplus$ charges $\vec{E}$
 $\ominus$ " "
 $\vec{E} = \frac{1}{q} \frac{\vec{F}}{q}$

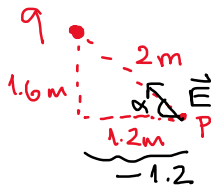


(c) = field x
are more
q increase

surrounding three different point charges. (a) A positive charge. (b) A negative

$$E = \frac{kq}{r^2}$$

ex) what's $\vec{E}$ at point P?



$$q = -8 \text{ nC}$$

$$E = \frac{kq}{r^2} = \frac{9 \times 10^9 \times 8 \times 10^{-9}}{2^2}$$

{ Force exerted on
a unit charge }

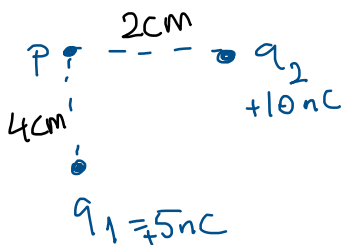
direction $\Rightarrow$ angle.

$$\alpha = \tan^{-1} \frac{1.6}{-1.2} = -53^\circ$$

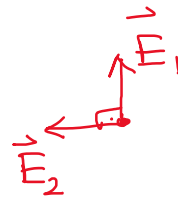
$$E = 18 \text{ N/C}$$



ex)

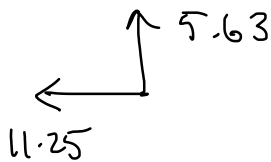


$$\vec{E}_P = ?$$



$$E_1 = \frac{9 \times 10^9 \times 10 \times 10^{-9}}{16 \times 10^{-4}} = 5.63 \times 10^4 \text{ N/C}$$

$\sum \vec{E} = \vec{E}_1 + \vec{E}_2$
superposition



$$E_2 = \frac{9 \times 10^9 \times 5 \times 10^{-9}}{4 \times 10^{-4}} = 11.25 \times 10^4$$

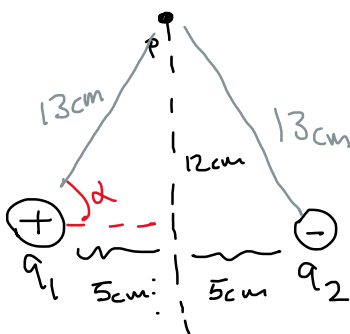
$$\sum \vec{E} = (5.63 \hat{j} + 11.25 \hat{i}) \times 10^4 \text{ N/C}$$

$$|\sum \vec{E}| = \sqrt{5.63^2 + 11.25^2} \quad ; \quad \theta = \tan^{-1} \left(\frac{E_y}{E_x} \right) = \tan^{-1} \left(\frac{5.63}{-11.25} \right)$$

$$= 12.6 \times 10^4 \text{ N/C}$$

$$= -26.6^\circ$$

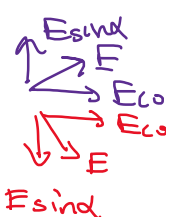
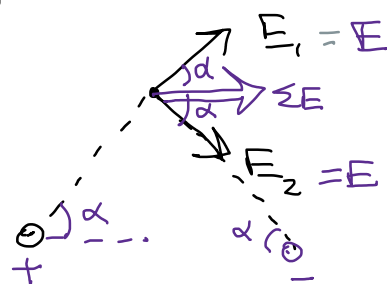
ex)



$$q_1 = +12 \text{ nC}$$

$$q_2 = -12 \text{ nC}$$

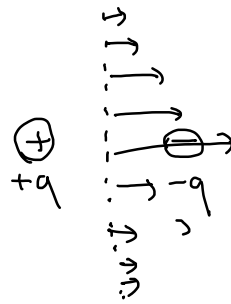
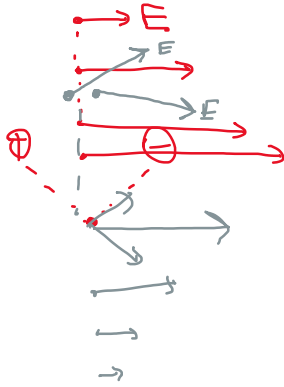
$$\vec{E}_P = ?$$



$$E_1 = \frac{kq}{r^2} = \frac{9 \times 10^9 \times 12 \times 10^{-9}}{13^2} = E_2 = 6.4 \times 10^3 \text{ N/C}$$

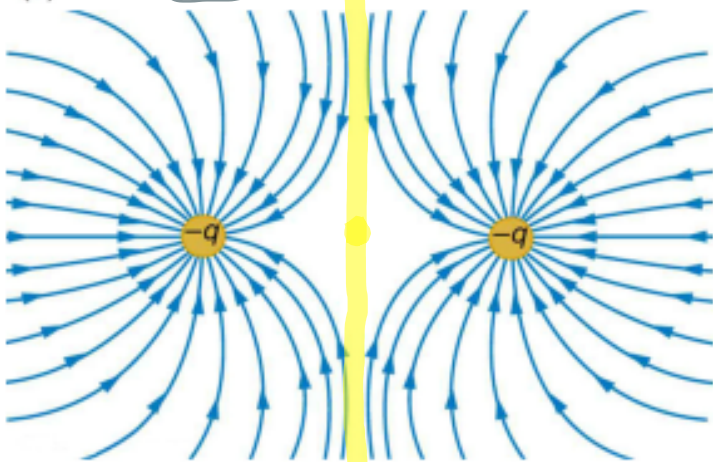
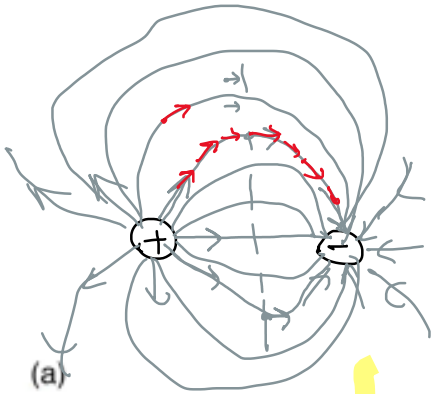
$$\sum E_P = 2E$$

$$13^+ \times 10^4$$



$$\sum E_p = 4.9$$

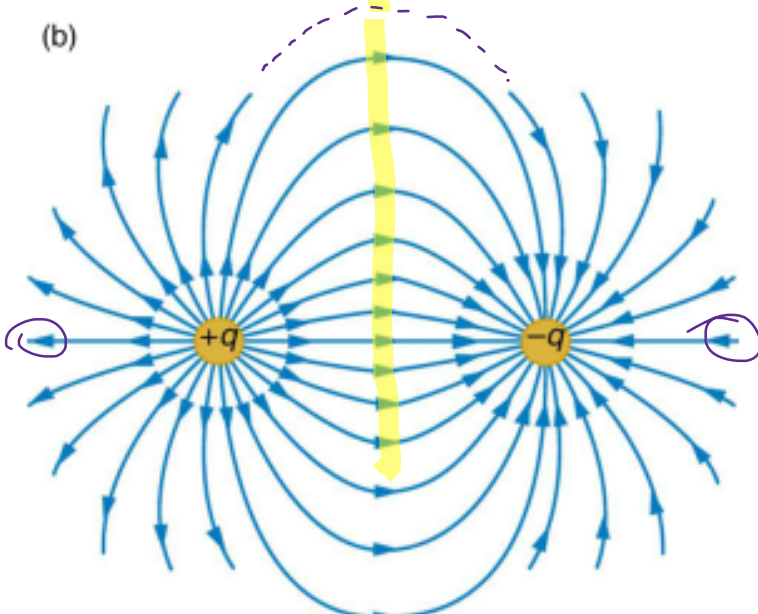
dipole $\begin{matrix} + & - \\ q & -q \end{matrix}$
 ↓
 molecules have
 dipole proper
 dipole



$$+2q$$



Symmetry:

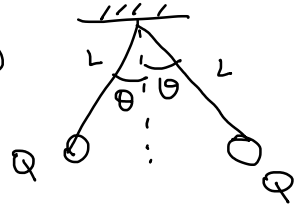


We use electric field lines to visualize and analyze electric fields (the lines are a pictorial tool, not a physical entity in themselves). The properties of electric field lines for any charge distribution can be summarized as follows:

1. Field lines must begin on positive charges and terminate on negative charges, or at infinity in the hypothetical case of isolated charges.
2. The number of field lines leaving a positive charge or entering a negative charge is proportional to the magnitude of the charge.
3. The strength of the field is proportional to the closeness of the field lines—more precisely, it is proportional to the number of lines per unit area perpendicular to the lines.
- 4.
5. The direction of the electric field is tangent to the field line at any point in space.
6. Field lines can never cross.

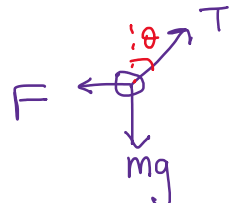
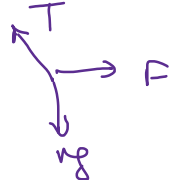
From <https://openstax.org/books/college-physics-2e/pages/18-5-electric-field-lines-multiple-charges>

ex)



$\theta = 5^\circ$ $L = 0.5\text{m}$ $m = 5 \times 10^{-3}\text{kg}$ $Q = ?$

FBD

$Q = \text{same charge}$

$T \cos \theta$
 $F \leftarrow T \sin \theta$
 mg

$a = 0$ $\Sigma F_x = 0$ $\Sigma F_y = 0$

$F = T \sin \theta$ $mg = T \cos \theta$

$\frac{F}{mg} = \tan \theta$

$F = mg \tan \theta$

$m \checkmark g \checkmark \tan \theta \checkmark$

$F = k \frac{q_1 q_2}{r^2}$



$L \sin \theta + L \sin \theta = r$

$F = \frac{9 \times 10^9 \times Q^2}{(2 \times 0.5 \times \sin 5^\circ)^2} = 5 \times 10^{-3} \times 9.8 \times \tan 5^\circ$

$Q = \left[\frac{5 \times 9.8 \times \sin^2 5^\circ \times \tan 5^\circ \times 10^{-3}}{9 \times 10^9} \right]^{1/2} \sim \sqrt{\frac{3.6 \times 10^{-6}}{10^9}} = \sqrt{36} = 6$

$\frac{k Q^2}{r^2} = mg \tan \theta \Rightarrow Q = \left[\frac{mg r^2 \tan \theta}{k} \right]^{1/2}$

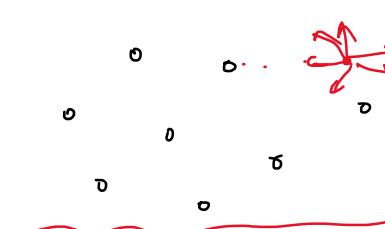
$Q = \left[\frac{mg 4L^2 \sin^2 \theta \tan \theta}{k} \right]^{1/2}$

✓ point charge

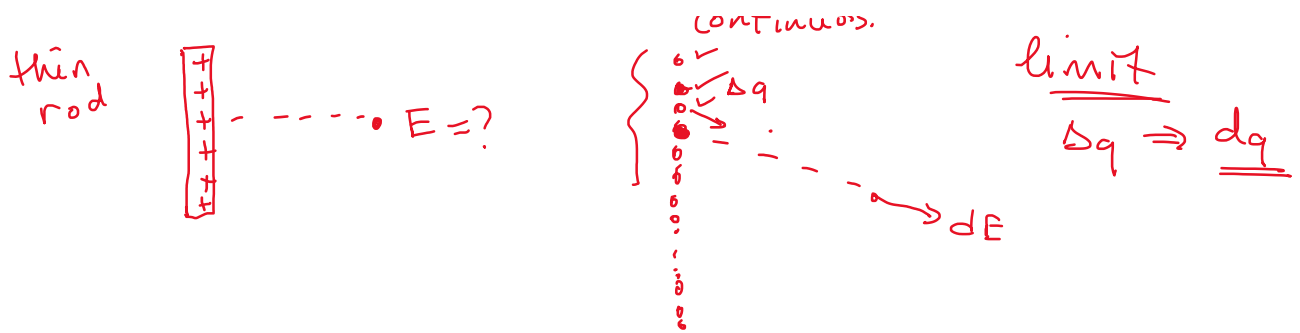


$$E = \frac{kq}{r^2}$$

multiple point charge.

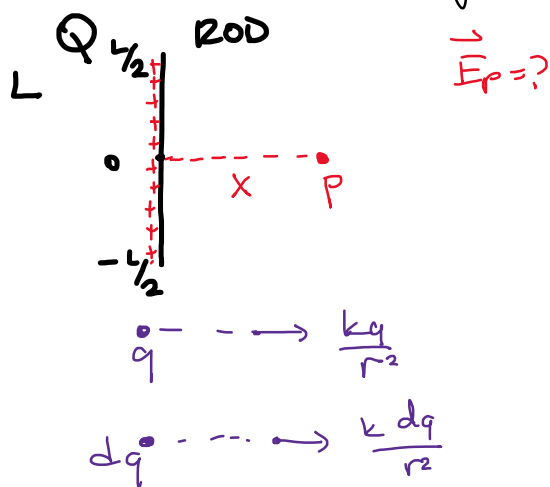


$$\Sigma \vec{E}_i = \Sigma \frac{k q_i}{r_i^2} \hat{r}_i$$

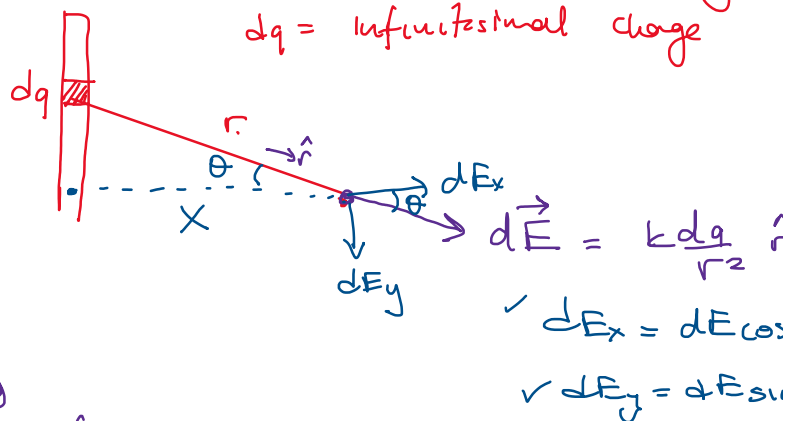


Electric field of a uniformly charged rod
discrete. continuous.

ROD = 1 dim object.



Divide the object infinitesimally so
 $dq =$ infinitesimal charge



$$\int k \frac{dq}{r^2} \cos \theta = E_x$$

$$\int k \frac{dq}{r^2} \frac{x}{r} = \int \frac{k dq x}{(x^2 + y^2)^{3/2}}$$

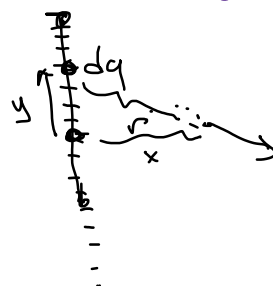
$$dq \leftrightarrow y$$

$$\int dE_x = E_x$$

$$\int dE_y = E_y$$

$$r = \sqrt{x^2 + y^2}$$

$$\cos \theta = \frac{x}{r} = \frac{x}{\sqrt{x^2 + y^2}}$$



uniformly charged $\Rightarrow$ charge density is constant $\frac{1}{2}$

$\lambda = \frac{Q}{L} = \frac{dq}{dy} \Rightarrow dq = \frac{Q}{L} dy$

$$E_x = \int \frac{k \lambda x}{(x^2 + y^2)^{3/2}} dy = \int_{-L/2}^{L/2} \frac{k \lambda x Q}{L} \frac{dy}{(x^2 + y^2)^{3/2}}$$

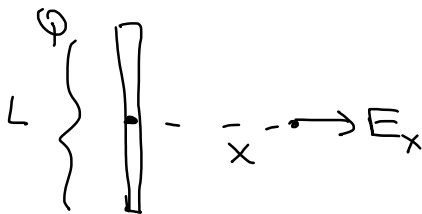
$$\frac{k \times Q}{L} \int_{-L/2}^{L/2} \frac{dy}{(x^2 + y^2)^{3/2}} \sim \int \frac{dy}{y^3} \text{ const.}$$

int. table

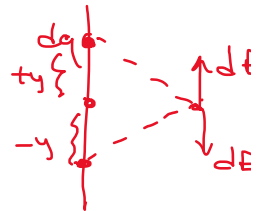
$$\int \frac{dy}{(x^2 + y^2)^{3/2}} = \left[\frac{1}{x^2} \frac{y}{(x^2 + y^2)^{1/2}} \right]_{-L/2}^{L/2}$$

$$\frac{k \times Q}{L} \frac{1}{x^2} \left[\frac{L/2}{(x^2 + L^2/4)^{1/2}} - \frac{-L/2}{(x^2 + L^2/4)^{1/2}} \right] = \frac{k \times Q}{x^2} \frac{L}{(x^2 + \frac{L^2}{4})^{1/2}}$$

$$E_x = \frac{kQ}{x} \frac{1}{\left[x^2 + \frac{L^2}{4}\right]^{1/2}} \quad E = \frac{kQ}{r^2} \sim k \frac{\text{charge}}{\text{length}^2}$$

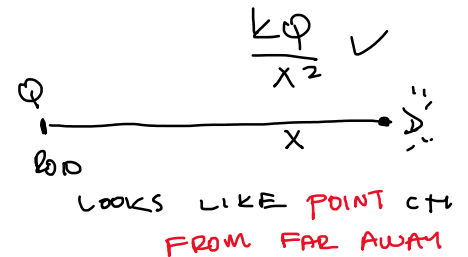


$$dE_y = ? \quad \int dE_y = 0$$



$$\left\{ \begin{array}{l} L \int_{-L/2}^{L/2} \frac{y}{\sqrt{x^2 + y^2}} dy \\ \frac{kQ}{xL} \left[\frac{y}{\sqrt{x^2 + y^2}} \right]_{-L/2}^{L/2} \end{array} \right\} \int dE_y \neq 0$$

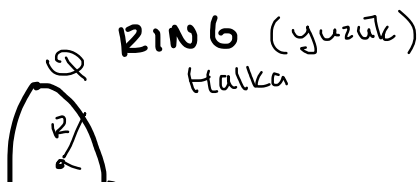
$$L \int_{-L/2}^{L/2} \frac{y}{\sqrt{x^2 + y^2}} dy \quad E_x = \frac{kQ}{x} \frac{1}{\left[x^2 + \frac{L^2}{4}\right]^{1/2}}$$

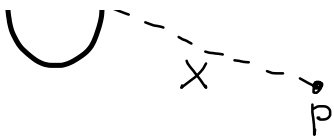


① $x \gg L \quad \frac{L}{x} \approx 0$

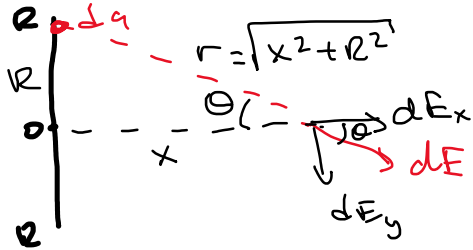
② $x \ll L$

$$\frac{kQ}{x} \frac{1}{\left[x^2 \left(1 + \frac{L^2/4}{x^2}\right)^{1/2}\right]^{1/2}} = \frac{kQ}{x} \frac{1}{x} = \frac{k}{x^2}$$





SIDE VIEW RING



$$dE = \frac{k dq}{r^2} = \frac{k dq}{x^2 + R^2}$$

$$dE_x = dE \cos \theta = \frac{k dq}{(x^2 + R^2)} \frac{x}{\sqrt{x^2 + R^2}}$$

$$\int dE_x = \int \frac{k x dq}{(x^2 + R^2)^{3/2}}$$

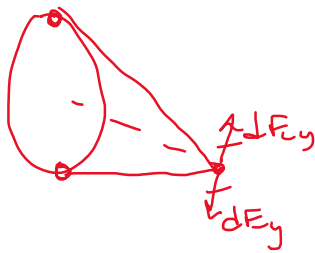
constant !!

$k, x, R = \text{const}$

$$\int \frac{k dq x}{r^{3/2}}$$

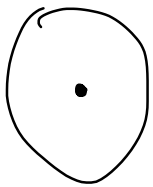
$$= \frac{k x}{(x^2 + R^2)^{3/2}} \underbrace{\int dq}_Q = \frac{k Q x}{(x^2 + R^2)^{3/2}} = E_x$$

* $\int dE_y \approx 0$



IF YOU ARE FAR AWAY
IT STAYS
 $x \gg R$

$$\frac{R}{x} \approx 0$$



AT THE CENTER

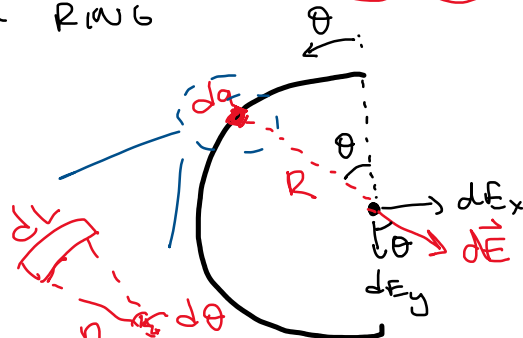
$$x=0 \quad \frac{k Q (0)}{(0^2 + R^2)^{3/2}} = 0$$

$$\frac{k Q x}{\left[x^2 \left(1 + \frac{R^2}{x^2} \right) \right]^{3/2}} = \frac{k Q}{x}$$

Half charged RING



$\vec{E}_0 = ?$



$$\int \frac{k dq}{r^2} \hat{r} = \int dE_x$$

$$\int dE_x = \int \frac{k dq}{R^2} \sin \theta$$

$$\text{charge density} = \frac{Q}{L} = \frac{Q}{\pi R} = \frac{dq}{R d\theta} = \lambda$$



$$\int dE_x = dE \sin \theta$$

$$\int dE_y = dE \cos \theta$$

$$\lambda = \frac{Q}{\pi R} = \frac{dq}{R d\theta} \Rightarrow dq = \frac{Q d\theta}{\pi}$$

charge density

$$dq = \lambda d\theta$$

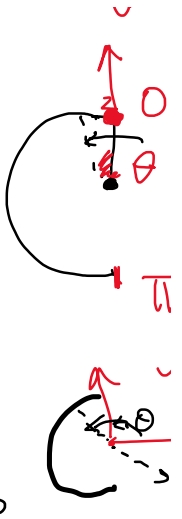
$$\lambda = \frac{dq}{R d\theta}$$

$$dL = R d\theta \text{ radian}$$

$$L_1 = R\theta_1$$

$$\int \frac{k dq}{R^2} \sin\theta = \int \frac{k Q}{\pi R^2} d\theta \sin\theta = \frac{k Q}{\pi R^2} \int_0^\pi \sin\theta d\theta$$

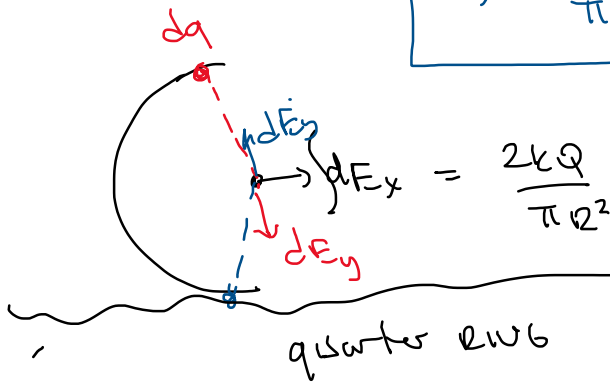
$$\left[-\cos\theta \right]_0^\pi$$



$$\frac{k Q}{\pi R^2} [-\cos\theta]_0^\pi = \frac{k Q}{\pi R^2} [-\cos\pi - (-\cos 0)]$$

$$= \frac{k Q}{\pi R^2} [1 + 1] = \frac{2kQ}{\pi R^2}$$

$$E_x = \frac{2kQ}{\pi R^2}$$



$$dE_y = \frac{k dq}{R^2} \cos\theta$$

$$= \frac{k Q}{\pi} \int_0^\pi \cos\theta d\theta$$

$$\left[\sin\theta \right]_0^\pi$$

$$E_x = \frac{k Q}{\pi R^2} [-\cos\theta]_{\theta_1}^{\theta_2}$$

$$E_y = \frac{k Q}{\pi R^2} [\sin\theta]_{\theta_1}^{\theta_2}$$

$$E = \frac{2}{\pi} \frac{kQ}{R^2}$$

$$\Rightarrow \frac{2kQ}{\pi R^2} + \frac{2kQ}{\pi R^2}$$

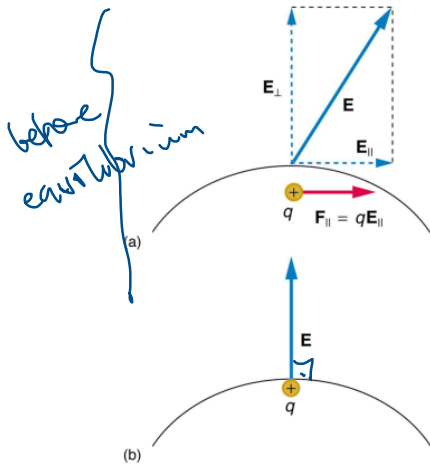
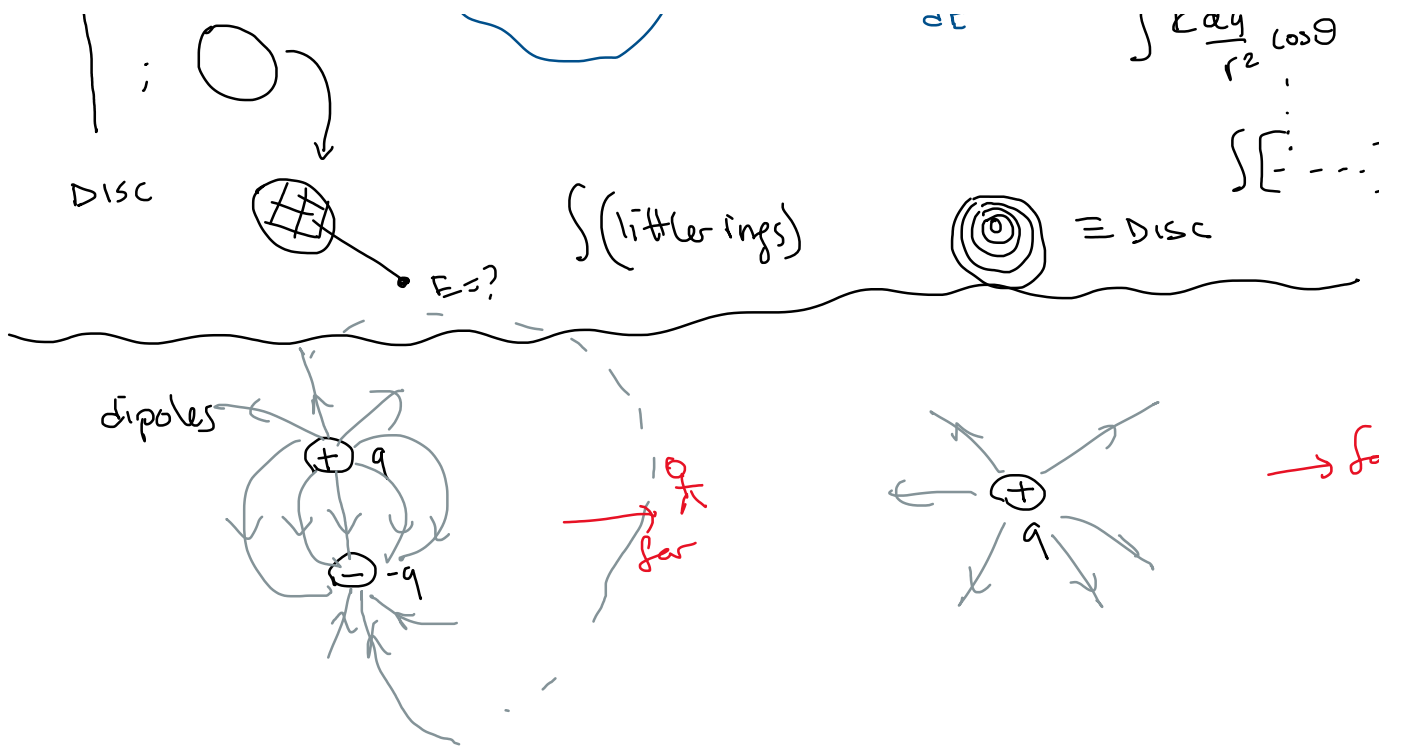


$$d\vec{F} = k \frac{dq}{r^2} \hat{r}$$

rod

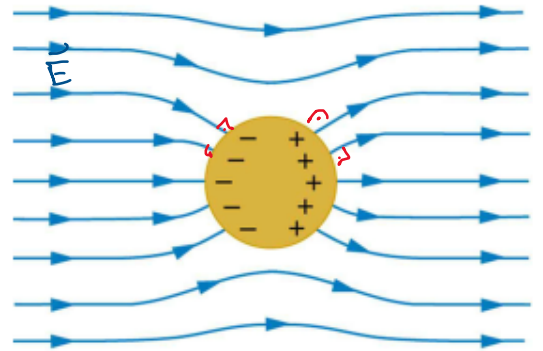
ring

$$dq \leftrightarrow r, x, y, z$$

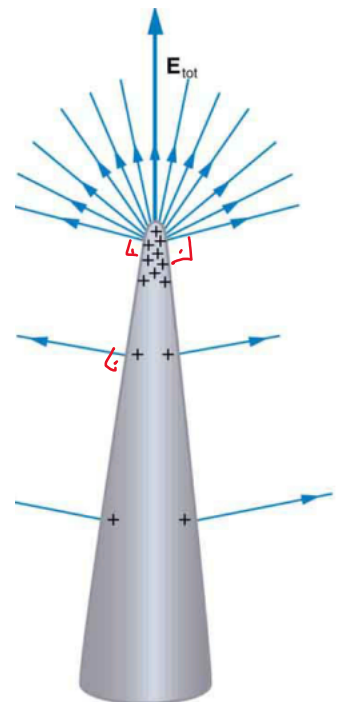
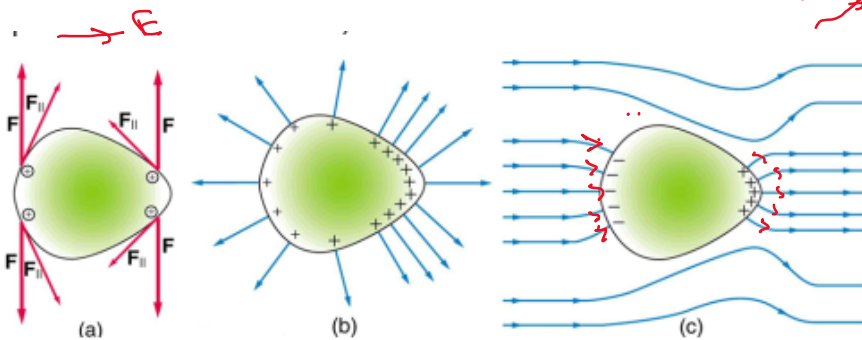


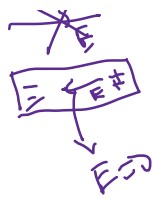
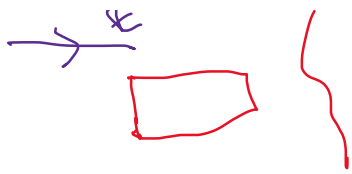
conductors have freely moving charge.

↓
move to equilibrium fast
⇓
electrostatic

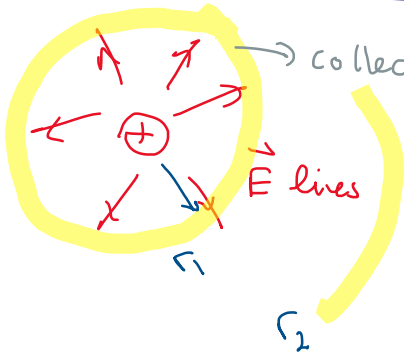


Irregular shape conductors





GAUSS LAW



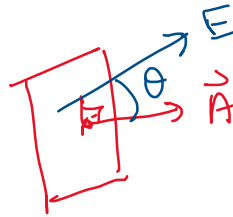
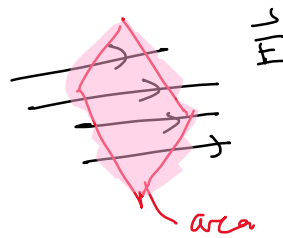
no of lines will be equal!

Electric field flux = $\Phi = (E \cdot \text{area})$
(phi)

$\Phi = \vec{E} \cdot \vec{A}$ FLUX
area

$\left\{ \begin{array}{l} E \uparrow \quad \phi \uparrow \\ A \uparrow \quad \phi \uparrow \end{array} \right\}$

Area



$\vec{E} \cdot \vec{A}$
dot product

$|\vec{E}| |\vec{A}| \cos \theta = \Phi$

$\Phi = \vec{E} \cdot \vec{A} \rightarrow 0$

$\Phi = 0$

$\Phi_{\text{max}} \quad \uparrow E \uparrow A \quad \uparrow F$
 $\Phi = 0 \quad \vec{E} \perp A \quad \vec{E} \perp A \quad \Phi$

$\Phi = \vec{E} \cdot \vec{A} = [?]$

point charge

$E = \frac{kq}{r^2}$

E field $\vec{F} = q\vec{E}$

$\left[\frac{N}{C} \right] = E$

$\Phi = \vec{E} \cdot \vec{A} = \left[\frac{N}{C} m^2 \right] = \left[\frac{Nm^2}{C^2} \right] \downarrow C$

unit analysis

$k = ?$

$\frac{kq}{r^2} = E = \left[\frac{N}{C} \right]$

$k = \frac{Nm^2}{C^2} = \frac{1}{4\pi\epsilon_0}$

$\Phi = kq$

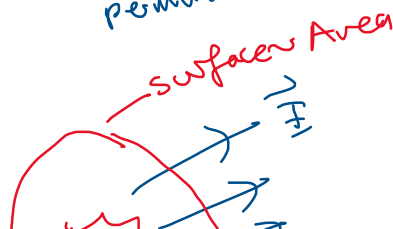
$= \frac{q}{\epsilon_0}$

$k = \frac{1}{4\pi\epsilon_0}$
Col. const

elec. permittivity

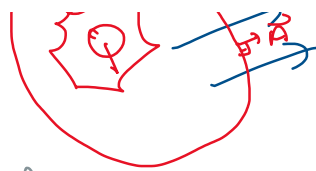
Gauss Law

$\Phi_{\text{net}} = \Phi = \vec{E} \cdot \vec{A} = \frac{q_{\text{in}}}{\epsilon_0}$



$$\Phi = \oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

closed system method = fines



q_{in} = total charge inside $\oint dA$ sphere

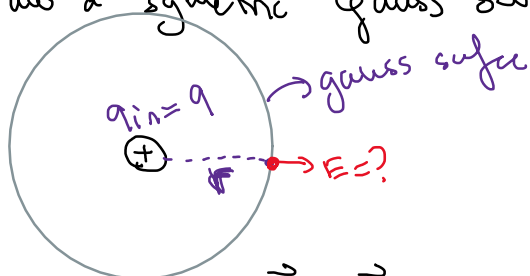
what is the $\vec{E}$ field of a point charge?

→ Gauss Law

$$\oplus q \cdots r \cdots \vec{E} = \frac{kq}{r^2} = \frac{q}{4\pi\epsilon_0 r^2}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

① draw a symmetric Gauss surface encloses the charge.



so that $\vec{E} \cdot d\vec{A}$ is easy and E is const. on the Gauss surface



$$\textcircled{2} \quad \vec{E} \cdot d\vec{A} \quad \int \vec{E} \cdot d\vec{A} = E dA \cos \theta \quad \oint E dA = \frac{q}{\epsilon_0}$$

$$\textcircled{3} \quad E \text{ const. on the surface} \quad \oint E \quad \therefore E \oint dA = \frac{q}{\epsilon_0}$$

$$\textcircled{4} \quad E \int dA = \frac{q}{\epsilon_0}$$

A = area of sphere

A
area of Gauss surface

$$E 4\pi r^2 = \frac{q}{\epsilon_0} \Rightarrow E = \frac{q}{4\pi\epsilon_0 r^2} \quad \checkmark$$



vs

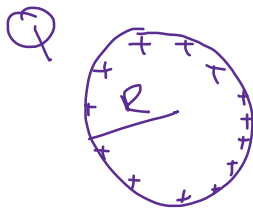


$$\Phi = \oint \vec{E} \cdot d\vec{A} = \frac{4q}{\epsilon_0}$$

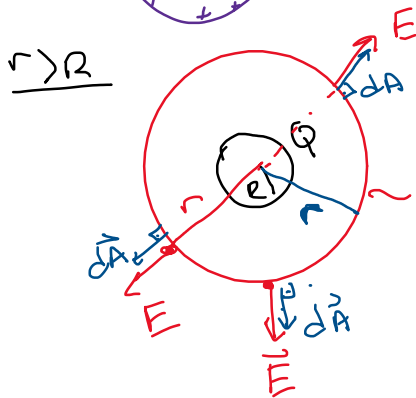
flux

11 gauss surface 44

charged spherical shell = charged metal sphere.



$$\vec{E}(r > R) = ? \quad \vec{E}(r < R) = ?$$



$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0} = \frac{Q}{\epsilon_0}$$

$\vec{A}$ is normal to the surf

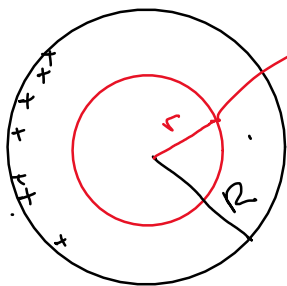
$$\oint E dA \cos 0^\circ = E \oint dA = \frac{Q}{\epsilon_0}$$

area of gauss. surf

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r} \quad (r > R)$$

$$E = 4\pi r^2 =$$

$r < R$
if metal sphere
all charge ON SURFACE



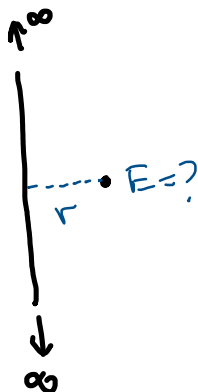
$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0} = 0$$

$$dA \neq 0$$

$$\int E \cdot dA = 0$$

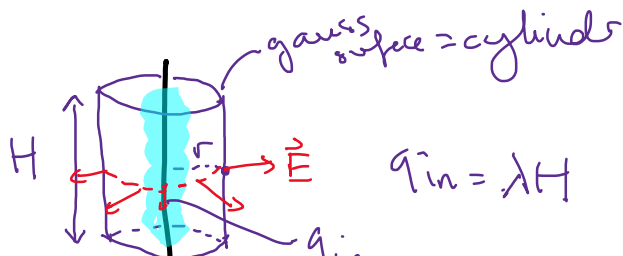
$$E = 0 \quad (r < R)$$

charged ROD; with infinite length; with λ = charge density.



$$\lambda = \frac{\text{charge}}{\text{length}} = \frac{Q}{L} = \text{const.}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$$



$$q_{in} = \lambda H$$

TOP VIEW

$$\oint \vec{E} \cdot d\vec{A} = \int \vec{E} \cdot d\vec{A} + \int \vec{E} \cdot d\vec{A} + \int \vec{E} \cdot d\vec{A}$$

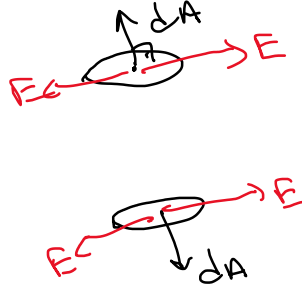
① ② ③

cylinder

$A_2 = 2\pi r H$

$$= \int \vec{E} \cdot d\vec{A}$$

① ③



$$\vec{E} \cdot d\vec{A} = E dA \cos 90 = 0$$

$$\oint \vec{E} \cdot d\vec{A} = 0 + E 2\pi r H + 0 = \frac{\lambda H}{\epsilon_0}$$

$$E = \frac{\lambda}{2\pi \epsilon_0 r}$$

Remember.

$$E = \int k \frac{dq}{r^2} = \frac{kQ}{x \sqrt{x^2 + \frac{L^2}{4}}}$$

$\lambda = \frac{Q}{L}$

$x \ll L$

$x \sqrt{\frac{L^2}{4} \left(1 + \frac{x^2}{L^2/4}\right)} \Rightarrow \left(\frac{xL}{2}\right) \left(1 + \dots\right)$

+q

point charge ✓

metal sphere



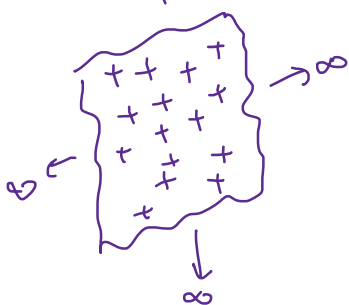
✓

∞ cap
R=0

✓

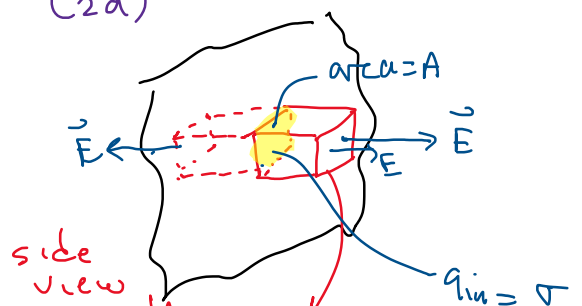
Infinitely large, charged surface with σ (sigma) charge density

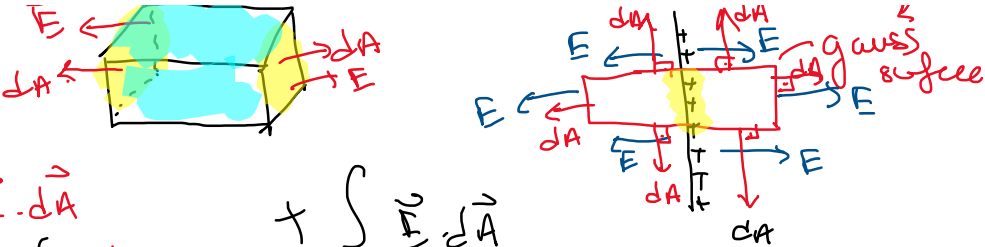
q_{∞}



$$\sigma = \frac{\text{charge}}{\text{area}} \quad (2d)$$

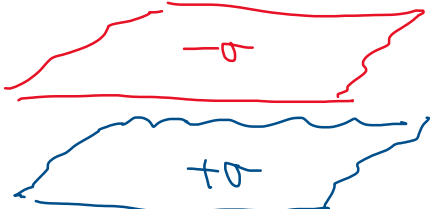
$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$$



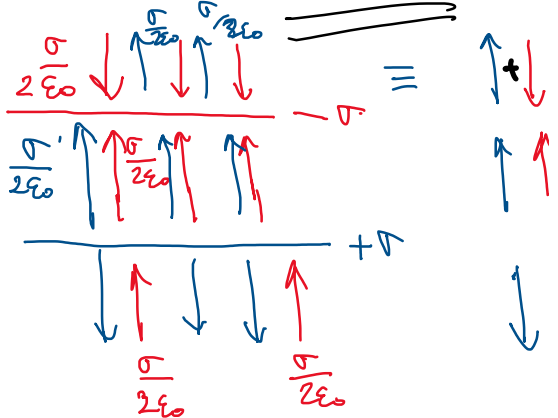


$$\int_{\text{sides}} \vec{E} \cdot d\vec{A} + \int_{\text{right}} \vec{E} \cdot d\vec{A} + \int_{\text{left}} \vec{E} \cdot d\vec{A} = \frac{q_{\text{in}}}{\epsilon_0} = \frac{\sigma A}{\epsilon_0}$$

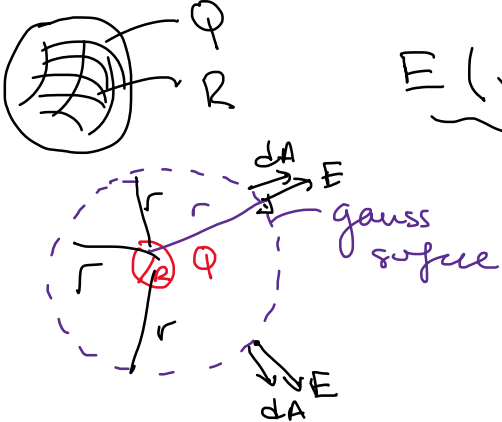
$\int_{\text{sides}} \vec{E} \cdot d\vec{A} = \int E dA \cos 90^\circ = 0$
 $\int_{\text{right}} \vec{E} \cdot d\vec{A} = E \int dA = EA$
 $\int_{\text{left}} \vec{E} \cdot d\vec{A} = E \int dA = EA$
 $EA + EA + 0 = \frac{\sigma A}{\epsilon_0} \Rightarrow 2EA = \frac{\sigma A}{\epsilon_0} \Rightarrow E = \frac{\sigma}{2\epsilon_0}$



Capacitor

$$E = \frac{\sigma}{\epsilon_0}$$


full, charged dielectric (insulator) sphere with ρ (rho) charge den



$r > R$

$$\int \vec{E} \cdot d\vec{A} = \frac{q_{\text{in}}}{\epsilon_0} = \frac{Q}{\epsilon_0}$$

$$E \int dA = \frac{Q}{\epsilon_0}$$

$$E 4\pi r^2 = \frac{Q}{\epsilon_0}$$

$$E(r > R) = \frac{Q}{4\pi\epsilon_0 r^2}$$

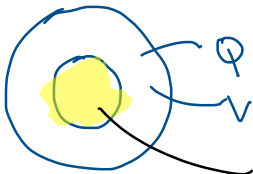
$r < R$



$r < R$

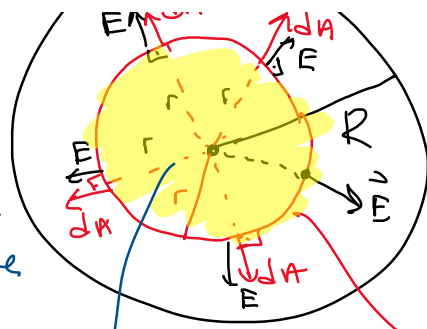
$$\rho(rho) = \frac{\text{charge}}{\text{volume}}$$

$$\rho = \frac{Q}{V}$$



$q_{in} =$
 q_{in}, V_{in}

$$q_{in} = \frac{Q \frac{4}{3} \pi r^3}{\frac{4}{3} \pi R^3} = \frac{Q}{R^3} \frac{r^3}{1}$$



gauss surface

$$\frac{q_{in}}{Q} = \frac{V_{in}}{V}$$

$$\oint E \cdot dA = \frac{q_{in}}{\epsilon_0}$$

$$\oint E \cdot dA \cos \theta = \frac{q_{in}}{\epsilon_0}$$

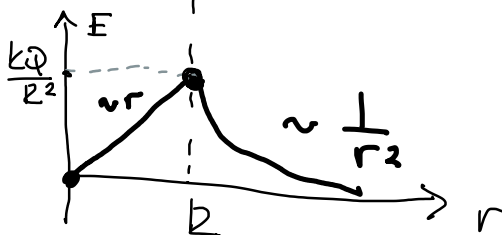
$$E \oint dA = \frac{q_{in}}{\epsilon_0}$$

$$\frac{Q}{V} = \rho = \frac{q_{in}}{V_{in}}$$

$$q_{in} = \frac{Q V_{in}}{V}$$

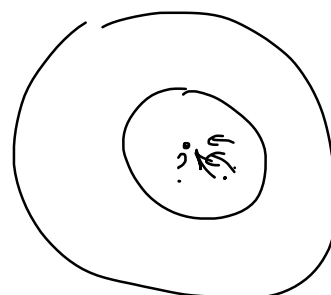
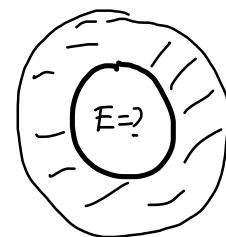
$$\rightarrow E 4\pi r^2 = \frac{Q}{\epsilon_0} \frac{r^3}{R^3} \Rightarrow E = \frac{Q r}{4\pi \epsilon_0 R^3} \quad (r < R)$$

$$\frac{Q R}{4\pi \epsilon_0 R^3}$$



$$\frac{Q r}{4\pi \epsilon_0 R^3}$$

$$\frac{Q}{4\pi \epsilon_0 r^2}$$



ρ, ρ inc. 0 0

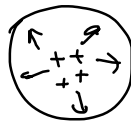
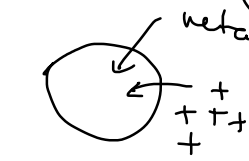
insulator

$$\rho = \frac{Q}{V} = \frac{Q}{\frac{4}{3}\pi R^3}$$

$$E = \frac{Q}{4\pi\epsilon_0 R^3} r$$

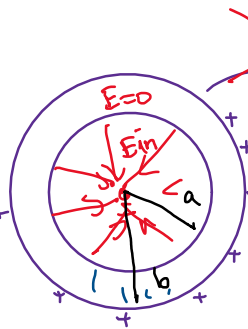
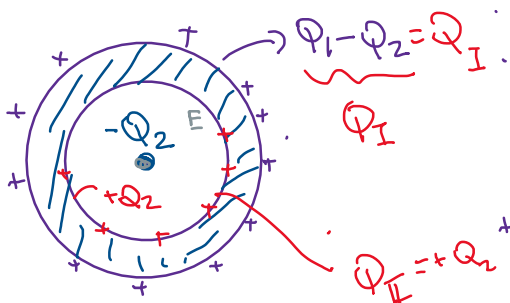
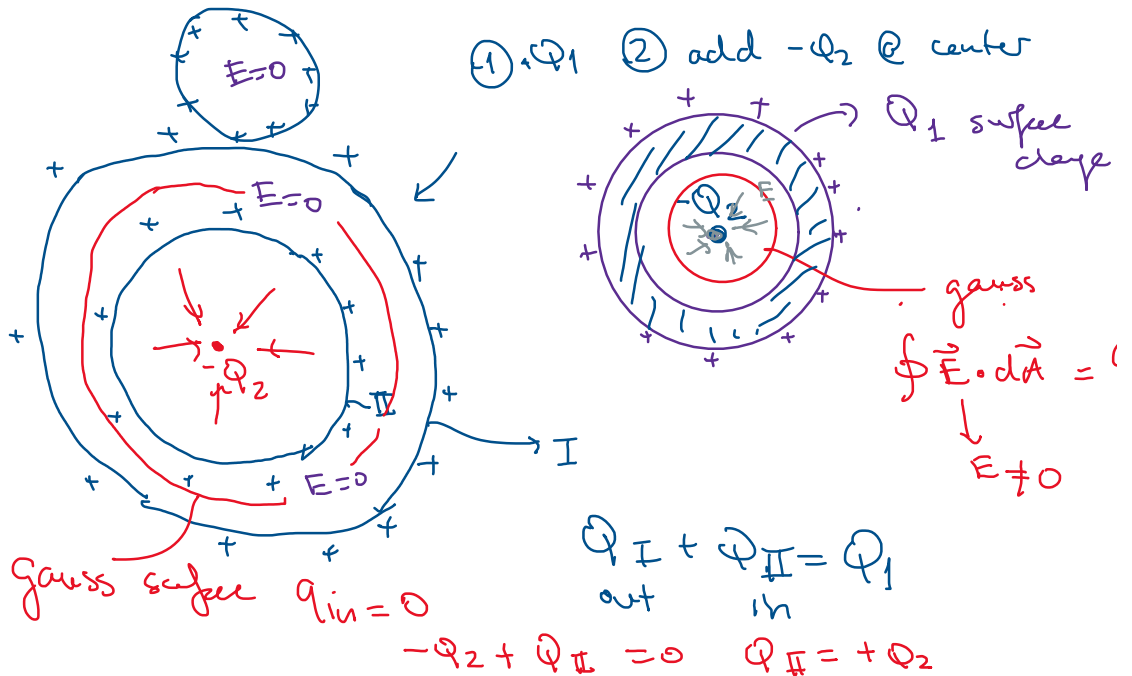
$$E = \frac{3\rho r}{\epsilon_0} = \text{const } r$$


conductors (metals) vs insulator.



all charges on the surface
time $< 1 \text{ ns} \sim 10^{-9} \text{ s}$

① $+Q_1$ ② add $-Q_2$ @ center



$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0} = Q$$

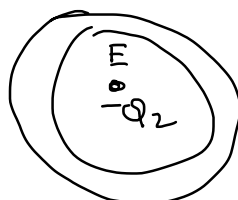
$$E_{in} = \frac{-Q_2}{4\pi\epsilon_0 r^2} \quad r < a$$

$$E_{out} = \frac{Q_1 - Q_2}{4\pi\epsilon_0 r^2} \quad r > b$$

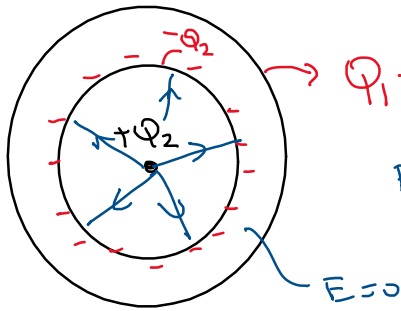
$$E = 0 \quad a < r < b$$



(=)



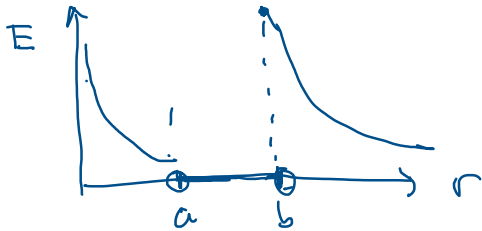
Initially $Q_{out} = +Q_1$ and we put $+Q_2$ at the c



$$Q_1 - (-Q_2) = Q_1 + Q_2$$

$$E(r < a) = \frac{Q_2}{4\pi\epsilon_0 r^2} ; E(a < r < b) = 0$$

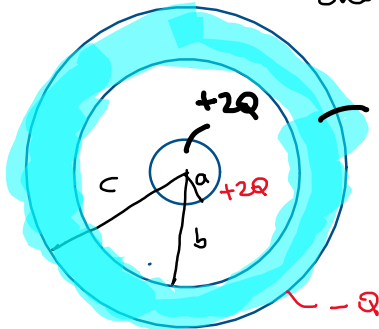
$$E(b < r) = \frac{Q_1 + Q_2}{4\pi\epsilon_0 r^2}$$



metal

charge of outer sphere is $-Q$

within this sphere; charge of $+2Q$ of radius a is placed.

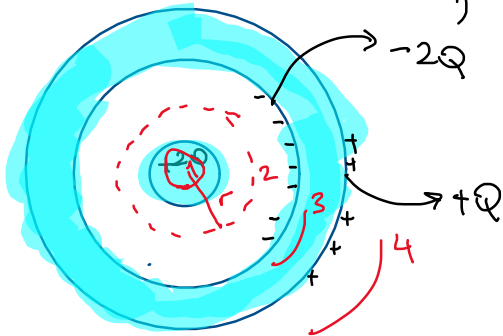


- 1) E ($r < a$)
- 2) E ($a < r < b$)
- 3) E ($b < r < c$)
- 4) E ($c < r$)



$$\oint \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$$

$$E 4\pi r^2$$



$$1) E ; (r < a) \quad q_{in} = 0 \quad E = 0$$

$$2) q_{in} (a < r < b) = +2Q \quad E = \frac{2Q}{4\pi\epsilon_0 r^2}$$

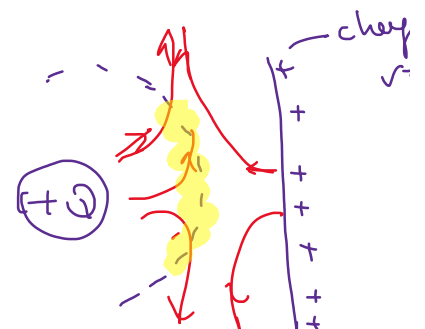
$$3) q_{in} (b < r < c) = 0 \quad E = 0$$

$$4) q_{in} (c < r) = +2Q - Q = +Q$$

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

$$\oint \vec{E} \cdot d\vec{A} = E 4\pi r^2 = \frac{+Q}{\epsilon_0}$$

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$



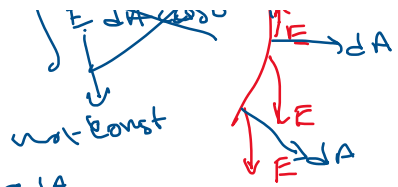
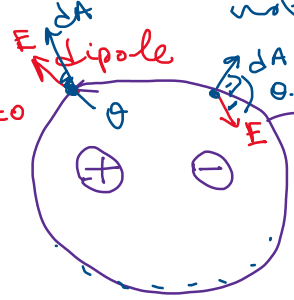
$$\oint \vec{E} \cdot d\vec{A}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0} = \Phi = \text{charge flux}$$

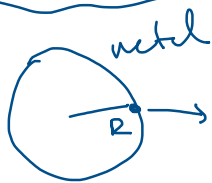
$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in} - q_{out}}{\epsilon_0} = 0$$

$$E=0; dA=0; \cos\theta=0$$

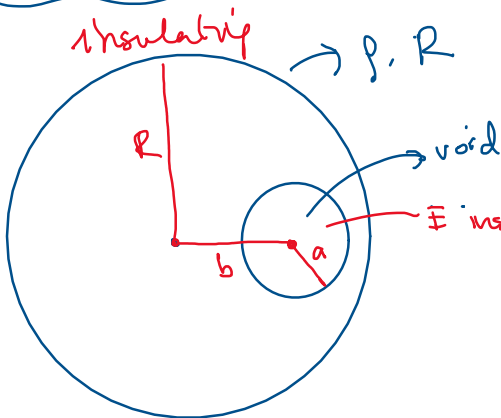
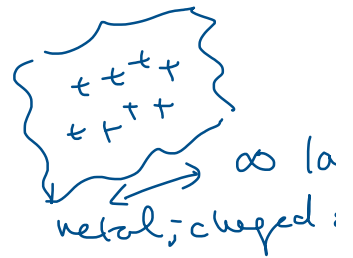
$\int \Rightarrow$



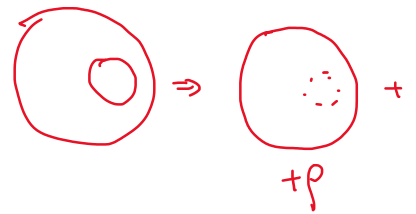
$$\vec{E} d\vec{A} = E dA \cos\theta + E dA \sin\theta$$



$$E = \frac{Q}{4\pi R^2 \epsilon_0} = \frac{\rho}{\epsilon_0} = \frac{V}{\epsilon_0}$$

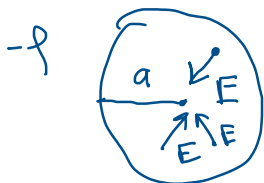
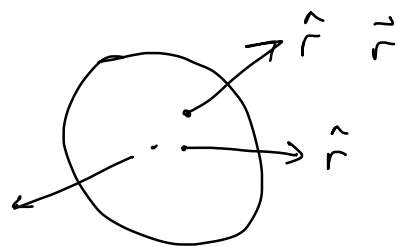


$$\text{void} = \text{charge } 0 = +Q + (-Q)$$

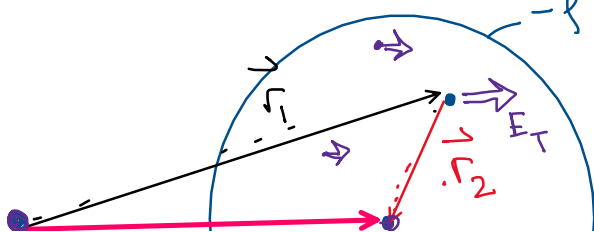
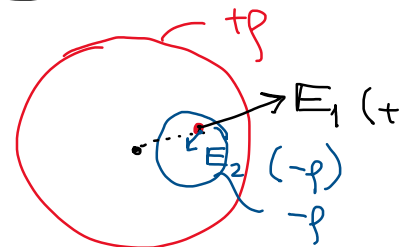


$$E(r < R) = \frac{3\rho}{\epsilon_0} r$$

$$\vec{E} = \frac{\rho}{\epsilon_0} \vec{r}$$



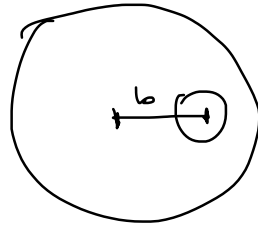
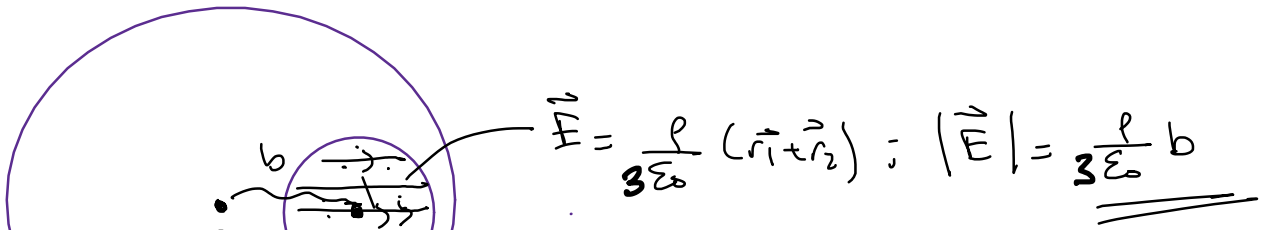
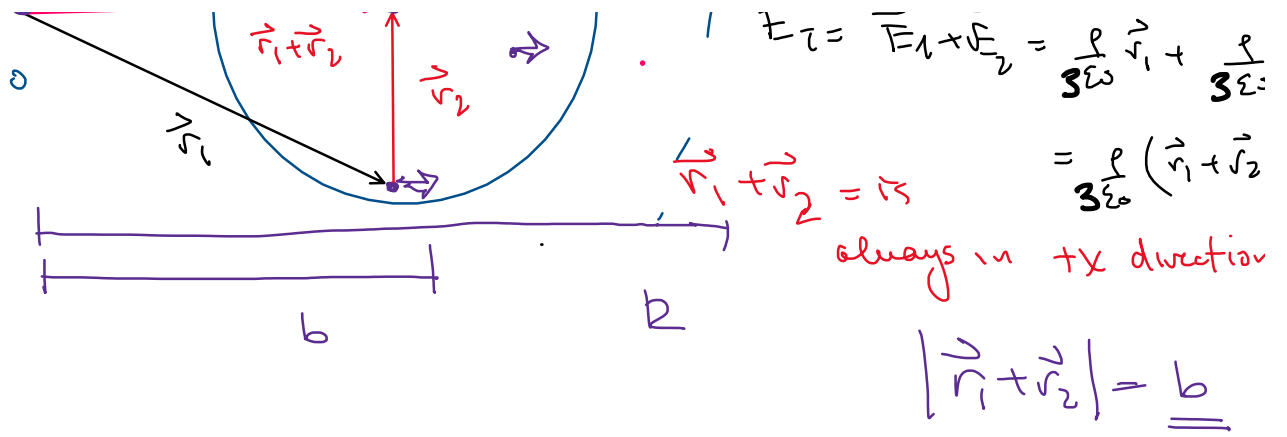
$$E(r < a) = -\frac{\rho}{\epsilon_0} \vec{r} = \frac{\rho}{\epsilon_0} (-\vec{r})$$



$$\vec{E}_1 = \frac{\rho}{\epsilon_0} \vec{r}_1$$

$$\vec{E}_2 = \frac{\rho}{\epsilon_0} \vec{r}_2$$

$$\vec{E}_1 + \vec{E}_2 = \vec{E}$$



$\Rightarrow$

